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Morphology-Tuned Pt_3Ge Accelerates Water Dissociation to Industrial-Standard Hydrogen Production over a wide pH Range

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The discovery of novel materials for industrial-standard hydrogen production is the present need considering the global energy infrastructure. A novel electrocatalyst, Pt_3Ge, which is engineered with a desired crystallographic facet (202), accelerates hydrogen production by water electrolysis, and records industrially desired operational stability compared to the commercial catalyst platinum is introduced. Pt_3Ge-(202) exhibits low overpotential of 21.7 mV (24.6 mV for Pt/C) and 92 mV for 10 and 200 mA cm^{-2} current density, respectively in 0.5 M H_2SO_4. It also exhibits remarkable stability of 15 000 accelerated degradation tests cycles (5000 for Pt/C) and exceptional durability of 500 h (@10 mA cm^{-2}) in acidic media. Pt_3Ge-(202) also displays low overpotential of 96 mV for 10 mA cm^{-2} current density in the alkaline medium, rationalizing its hydrogen production ability over a wide pH range required commercial operations. Long-term durability (>75 h in alkaline media) with the industrial level current density (>500 mA cm^{-2}) has been demonstrated by utilizing the electrochemical flow reactor. The driving force behind this stupendous performance of Pt_3Ge-(202) has been envisaged by mapping the reaction mechanism, active sites, and charge-transfer kinetics via controlled electrochemical experiments, ex situ X-ray photoelectron spectroscopy, in situ infrared spectroscopy, and in situ X-ray absorption spectroscopy further corroborated by first principles calculations.

1. Introduction

The ultimate dream of the human kind in the current scenario is to use the terms "fuel" and "combustion" without having to encounter any carbon dioxide (CO_2) or carbonaceous compounds. To date, the utilization of nonrenewable energy resources (like coal, petroleum, and natural gas) is exceeding the use of renewable ones like wind, water, biopower, and solar. The enormous growth of population has led to extensive consumption of these fossil fuels[1] and release of noxious CO_2 gas that has caused global warming, hiking the temperature at an alarming rate. Hydrogen (H_2) gas, in its molecular form, is the cleanest fuel known to date because its only combusted product is benevolent water.[2] Hydrogen can be used as a fuel in proton exchange membrane fuel cells (PEMFCs) for various routine applications,[3] as a reductant in the utilization of greenhouse gas CO_2,[4] as a raw material in the production of several chemicals using hydrogenation processes (e.g., ammonia synthesis,[5] methanol production[6]), as a direct fuel (e.g., in rocket[7]), in metallurgical industries,[8] semiconductor manufacturing,[9] pharmaceuticals,[8] and any other sectors having concerns related to energy and environment. However, large-scale H_2 production is majorly dependent on steam reforming of fossil fuels (about 96%),[10] and the rest 4% by water electrolysis.[11–17] Environmentally benign process, water electrolysis needs an exponential growth for the development of a powerful H_2 generator. Noble metals (platinum, palladium, and ruthenium[18]), especially platinum, are highly

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electroactive for hydrogen generation via water splitting attributed to their favourable electronic structure and optimum energy for hydrogen adsorption and desorption processes. But, extensive commercialization of water electrolysis process is not yet completely achieved owing to high cost of Pt. These comerciallization hurdles of water electrolysis process can be overcome either by reducing the cost of the catalyst or improving the perfromance or combining both. Even after a long scientific journey of synthesizing non-Pt based electrocatalysts, none of them could set foot in the performance region of Pt.

Extensive research activities are being carried out to maximize the catalytic activity of Pt along with decreased mass loading.[19] The hydrogen evolution reaction (HER) activity of Pt can be tuned by alloying with other transition elements,[20–22] formation of heterostructures,[23–25] core–shell design,[25–27] substituting with carbides[28,29] or nitrides or phosphides[30] and even generation of single-atom catalysts.[16,31–34] Despite showing promising catalytic performance, these materials were found to be incapable of fulfilling one or the other parameters required for commercial operation. There are five critical parameters (hereafter coined as Fabulous Five (Fab-Five)) that dictate HER activity of a catalyst: (1) onset potential, (2) current density, (3) Tafel slope, (4) stability in wide pH range, and (5) industrial-standard durability. However, discovering a material holding all these parameters together is a challenging task. There are several strategies used to develop materials, for example, combining Pt with 3d transition metals (TMs) or p block elements. This can alter the 5d electron occupancy of Pt and Pt–Pt interatomic distance, tuning d-band center and modifying the electronic structure.[35] The electronic structure modification improves the charge transfer kinetics and intermediate stability on the surface thereby lowering the activation barrier of the reaction.

Although several approaches have been explored,[36–40] there still lacks clarity in which atomic level modifications alter the electronic structure enhancing the HER activity close to the theoretical value. One such concept is to tune the arrangement of atoms within the crystal lattice, which can be achieved by the synthesis of ordered intermetallic compounds. Nevertheless, the situation can be even more complicated as the atomic packing at different crystallographic planes may alter the HER activity. It is worth mentioning that tuning a specific surface for catalysis at the atomic level is extremely difficult. There are a few handpicked numbers of works which proposed this concept by different synthetic strategies.[41–43] In this work we have introduced a novel concept, where we selectively tuned a surface favourable for HER activity close to the theoretical target. To achieve enhancement in Fab-Five parameters exceeding that of the state-of-the-art commercial Pt, we have selected an unexplored intermetallic compound Pt_3Ge (tetragonal system and space group: *I*4/*mcm*) (**Figure 1**a). Selective exposure of facets is typically done by the use of capping agents, whereas our work eliminates surfactants from the synthesis completely classifying it as a green approach. Our work highlights the in situ formation of a secondary phase which facilitates the exposure of an HER active facet. To the best of our knowledge, this new

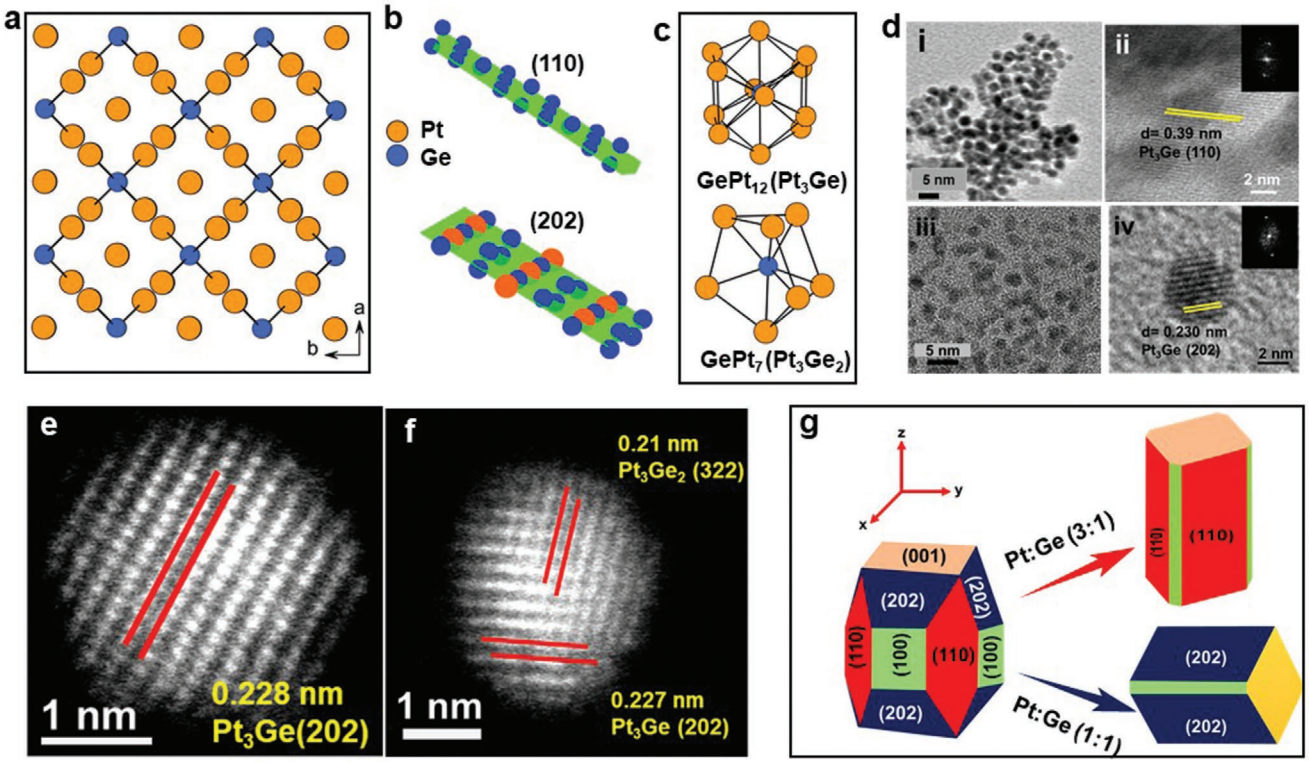

Figure 1. a,b) Crystal structure of Pt_3Ge (a) and (110) and (202) facets of Pt_3Ge (b). c) The coordination environment of Pt and Ge in Pt_3Ge and Pt_3Ge_2. d) Transmission electron microscopy (TEM) images of Pt_3Ge-(110) (i) and Pt_3Ge-(202) (iii) nanoparticles. Scale bars: 5 nm for (i) and (iii). High-resolution TEM (HR-TEM) images of Pt_3Ge-(110) (ii) and Pt_3Ge-(202) (iv) nanoparticles. FFT images shown as inset figures in (ii) and (iv). e,f) HAADF-STEM images of Pt_3Ge-(202). g) Schematic of the synthesis illustrating tuned formation of Pt_3Ge-(110) and Pt_3Ge-(202).

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catalyst is the first HER electrocatalyst which surpasses all Fab-Five parameters mentioned earlier. Using several controlled experiments including synthesis, electrochemical studies, spectroscopic measurements, operando FT-IR, X-ray absorption (XAS), and density-fucntional theory (DFT) calculations, the driving force behind the stupendous performance of this unique material is being established. The industrial level current density (>500 mA cm^{-2}) has been achieved by the meticulous optimization of the flow reactor and long-term durability (>75 h) of the HER has been obtained even in the sluggish alkaline media (1 M KOH) for practical utilization.

2. Results and Discussion

Noble-metal-based compounds having T_3X (T = Noble Metal; X = transition metal/p-block elements) composition have been known for superior electrochemical performance.[44,45] Partial replacement of Pt atoms by Ge is a highly cost-effective endeavour since the price of Ge is much lesser than that of Pt. Germanium is reported for its tendency to form hydrides[46] in acidic media and promote the HER when incorporated with other active noble metals like ruthenium[47] and palladium.[48] Due to the stable formation of GeH[49] and GeH_2,[50] there is an expected enhancement of reaction kinetics of the HER toward making it Volmer–Tafel mechanism resembling Pt. Nonsimilar adjacent atoms can lead to stronger heteroatomic interactions facilitating rapid charge transfer and optimal Metal-H bond strengths.[51] To substantiate this, we have simulated (110) and (202) crystallographic planes of Pt_3Ge using DFT caculations. The selection of these planes has been done based on the fact that (110) has only Pt atoms (equivalent to Pt metal) and (202) has an ordered arrangement of Pt and Ge atoms (Figure 1b). The preliminary calculations including the formation energy, density of states (DOS), and free energy of H* adsorption provided a hint of (202) being the active surface for the HER.[52]

We have generated different crystallographic planes in Pt_3Ge selectively using different synthetic conditions (see Figure S1 and Table S1 in the Supporting Information for detailed synthetic strategies). Figure 1c shows the cuboctahedron and bicapped-pentagon co-ordination environment of Ge in tetragonal Pt_3Ge and orthorhombic Pt_3Ge_2 phases, respectively. Ge is relatively less coordinated in Pt_3Ge_2 phase than in Pt_3Ge phase. Thus, it can be interpreted that Pt_3Ge is more stable phase compared to Pt_3Ge_2. The facet-selective synthetic strategy for the crystal growth of Pt_3Ge with (202) (named as Pt_3Ge-(202)) and (110) (Pt_3Ge-(110)) phases is schematically shown in Figure 1g. As mentioned in the Supporting Information (synthesis details), lesser Pt precursor input resulted in the formation of more active facet (202) of Pt_3Ge as the exposed plane. Figure S3 (Supporting Information) shows wide region HRTEM images of Pt_3Ge-(202) and Pt_3Ge-(110).

In order to validate the powder XRD analysis on the anisotropic crystal growth, transmission electron microscopy (TEM) images have been taken on different samples. As seen in Figure 1d, Pt_3Ge formed nanospheres (Figure 1d(i),(iii)) with an approximate particle size of 2.5 and 3.3 nm for Pt_3Ge-(202) and Pt_3Ge-(110), respectively (Figure S2, Supporting Information). High-resolution TEM (HR-TEM) image of Pt_3Ge-(110) nanoparticles shows the most exposed facet being (110) plane with d-spacing around 0.39 nm (Figure 1d(ii)). Whereas, HRTEM of the sample with higher Ge (Figure 1d(iv)) shows the existence of Pt_3Ge phase exposing (202) plane with *d*-spacing of 0.23 nm for Pt_3Ge. Figure 1e,f shows the HAADF-STEM images of Pt_3Ge-(202) with both exposed facets of (202) facets of Pt_3Ge and (322) factes of Pt_3Ge_2. Figure S4 (Supporting Information) shows the HAADF-STEM images of Pt_3Ge-(110) nanoparticles. Figure S5 (Supporting Information) shows the low maginification HAADF-STEM images of Pt_3Ge-(202).

The presence of the orthorhombic Pt_3Ge_2 phase disrupts the crystal growth along the (110) direction and triggers the selective formation of the facet (202) of Pt_3Ge, which is in excellent agreement with the HRTEM analysis. In this design, Pt_3Ge_2 phase can be termed as the "sacrifical phase," which is later self-destroyed during the electrochemical HER process. Scanning electron microscopy (SEM) (Figure S6, Supporting Information) depicts the morphology and point energy-dispersive X-ray spectra (EDX) (Table S2, Supporting Information) confirm the expected elemental composition and distribution, which is in agreement with the Rietveld refinement (Table S1, Supporting Information). Inductively coupled plasma-optical emission spectrometry (ICP-OES) further confirms the bulk elemental composition (Table S3 and Figure S10, Supporting Information).

2.1. Hydrogen Production

In order to validate our proposed concept, we have performed the HER in both acidic and alkaline media (wide pH range).[53] **Figure 2**a shows the linear sweep voltammograms (LSVs) of Pt_3Ge-(110) and Pt_3Ge-(202) in acidic condition (0.5 M H_2SO_4) compared with commercial 20% Pt/C. The overpotential corresponding to 10 mA cm^{-2} (h_{10}) current density is 34, 21.7, and 24.6 mV, respectively, for Pt_3Ge-(110), Pt_3Ge-(202), and Pt/C (Table S4, Supporting Information). LSV data for Pt_3Ge-(202) and 20% Pt/C taken using rotating disk electrode (RDE) shows current density up to 2000 mA cm^{-2} at –0.6 V vs RHE (Figure 2b) with a remarkably low overpotential of 92 and 102 mV for 200 mA cm^{-2} current density (inset in Figure 2b) for Pt_3Ge-(202) and 20% Pt/C, respectively. The Tafel slope observed is similar for all the catalysts (Figure 2c) suggesting Volmer–Tafel mechanistic pathway of the HER[54,55] with faster reaction kinetics in the case of Pt_3Ge-(202). Pt_3Ge-(202) shows no degradation even after 15 000 ADT cycles (Figure 2d), whereas Pt_3Ge-(110) degraded by 9 mV after similar duration (Figure S8, Supporting Information). In contrary, the activity of 20% Pt/C degraded within 5000 ADT cycles (Figure S8c, Supporting Information). Additionally, Pt_3Ge-(202) also exhibited a remarkable durability of 500 h chronoamperometry (CA) at 10 mA cm^{-2} (Figure 2e), which is one of the highest reported durability for any material having the onset potential and overpotential better than Pt/C, till date.[40] Impedance studies were done for both the catalysts and compared with 20% Pt/C (Figure S10, Supporting Information). The charge transfer resistance for Pt_3Ge-(202) is 10 Ω which means faster charge transfer to the active sites where reactants are adsorbed.

To evaluate the commercial viability of the catalysts in multiple operational conditions,[56–60] the activity has been checked

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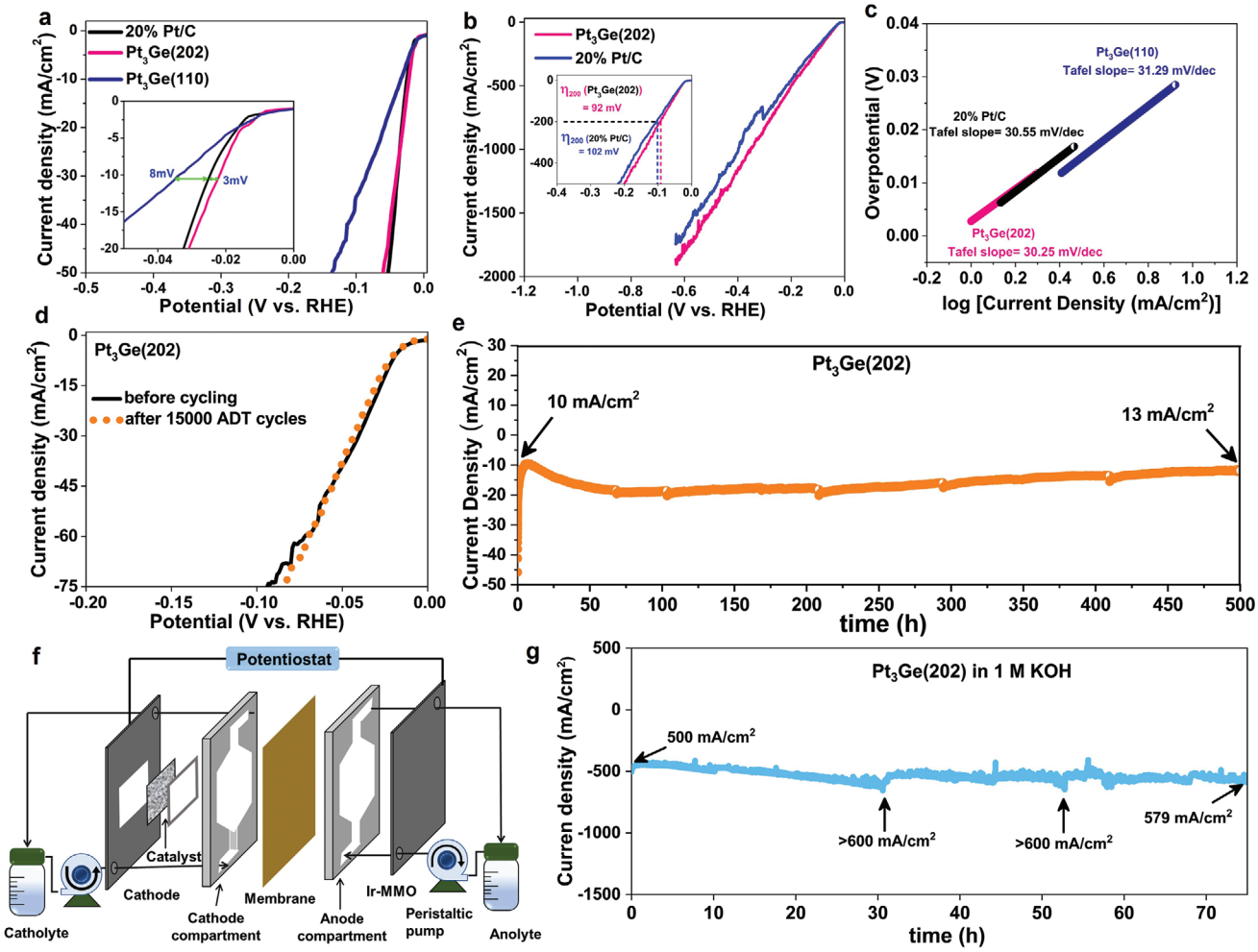

Figure 2. Electrochemical HER studies. a) Comparison of HER polarization curve (LSVs) of Pt_3Ge-(110), Pt_3Ge-(202) and 20% Pt/C, (ir-corrected). b) LSVs of Pt_3Ge-(202) and 20% Pt/C using rotating disk electrode (RDE) at 2000 rpm in 0.5 M H_2SO_4, (ir-corrected). c) Tafel slopes of Pt_3Ge-(110), Pt_3Ge-(202), and 20% Pt/C. d) LSVs of Pt_3Ge-(202) before and after 15 000 ADT cycles, respectively, in 0.5 M H_2SO_4. e) Chronoamperometric (CA) study for Pt_3Ge-(202) for 500 h at 10 mA cm^{-2} current density in 0.5 M H_2SO_4. f) Schematic of flow cell set-up for doing HER in 1 M KOH solution, and g) CA study for Pt_3Ge-(202) for 75 h at 500 mA cm^{-2} current density in 1 M KOH medium.

in the alkaline media (0.5 M KOH) as well. Pt_3Ge-(202) catalyst achieves 10 mA cm^{-2} current density at only 96 mV which is on par with the state-of-the-art catalyst 20% Pt/C and significant stability up to 5000 cycles with no degradation in h_{10} value (Figure S11, Supporting Information). Obtained Tafel slope value of 51.86 mV dec^{-1} is even smaller than that of 20% Pt/C (Figure S11b, Supporting Information). This signifies faster HER kinetics compared to Pt/C even in alkaline condition, which is also portrayed in the LSV where Pt_3Ge-(202) has lower overpotential compared to 20% Pt/C at the entire current density region. Figure S11d,e (Supporting Information) shows the activity of the catalysts in neutral medium (1 M PBS as neutral medium) and the stability of the catalyst upto 5000 ADT cycles was achieved. The catalyst Pt_3Ge-(202) exhibited 10 mA cm^{-2} at overpotential of 60 mV which is lesser than 20% Pt/C. Thus, the catalyst proves to be an efficient HER electrocatalyst in wide pH range.[61] To achieve industrial level current density, micro flow cell setup was optimized and alkaline HER in 1 M KOH medium was conducted. The schematic of the flow cell setup during the HER is shown in Figure 2f and Figure S12 (Supporting Information). Interestingly, the catalyst Pt_3Ge-(202) shows very high durability for 75 h when done at 500 mA cm^{-2} current density (Figure 2g). Figure S13 (Supporting Information) shows the LSV before and after 75 h CA in 1 M KOH medium using flow cell set-up.[62,63] Figure S14 (Supporting Information) shows the durability of 1 M KOH medium HER by 20% Pt/C. This is very clear that Pt_3Ge-(202) is much more durable and active than 20% Pt/C.[64–66] Cyclic voltammograms in the non-Faradaic region (0.25–0.85 V vs RHE) during the activation process of both catalysts are presented in **Figure 3**a,b, where a notable difference in Ge leaching during the oxidation sweep is observed. This can be correlated to the difference in the exposed crystallographic planes as ordered arrangement of Pt and Ge atoms are present exclusively on (202) facet of Pt_3Ge-(202) (Figure 1b). Moreover, as shown in Figure 1c, Ge is coordinated by 12 Pt atoms in Pt_3Ge and surrounded by 7 Pt atoms in Pt_3Ge_2. Due to it being less coordinated by Pt, Ge in Pt_3Ge_2 is more prone to get leached. These two factors drive the selective dissolution of

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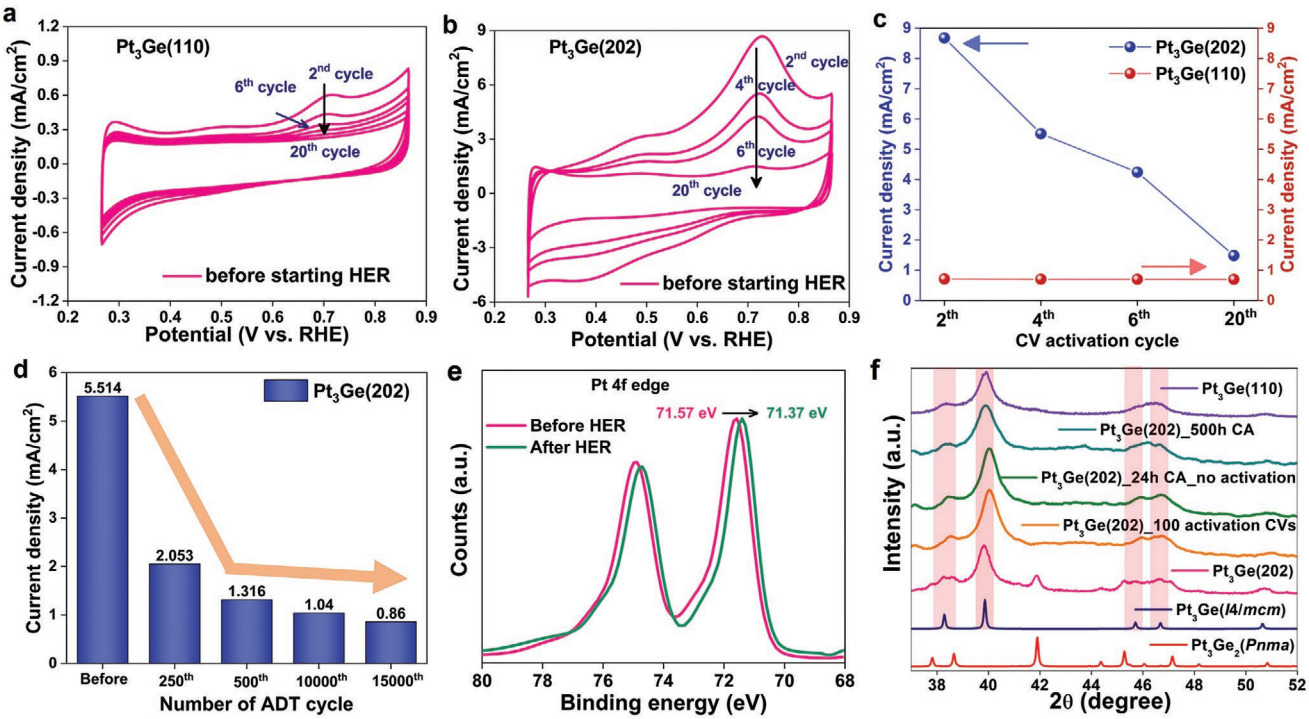

Figure 3. Electrochemical control experiments. a) Activation CVs of Pt_3Ge-(110) after each cycle. b) Activation CVs of Pt_3Ge-(202) after each cycle. c) Comparison of current density corresponding to Ge leaching for both catalysts during activation CVs as in (a) and (b) for Pt_3Ge-(202) and Pt_3Ge-(110). d) Current density comparison for different activation CVs before and after ADT cycles in HER potential range of Pt_3Ge-(202). e) Post-HER Pt 4f XPS spectra of Pt_3Ge-(202). f) Post-HER PXRD pattern comparison for Pt_3Ge-(202).

Ge from the Pt_3Ge_2 phase which leads to the operando disruption of Pt_3Ge_2, leading to the formation of pure Pt_3Ge phase.

With increasing cycles, the peak intensity of CV curve corresponding to Ge oxidation decreases (Figure 3a,b), indicating majority of the Ge atoms at the surface are leached in the first few cycles. The absence of any Ge oxidation peak at the 20th cycle confirms no further dissolution of Ge. As seen in Figure 3c (derived from Figure 3a,b), Ge leaching process is predominant in the case of Pt_3Ge-(202) with a substantial drop of current density from 9 to 1 mA cm^{-2}, which is negligible in the case of Pt_3Ge-(110). However, after a prolonged period of HER CVs, leached Ge redeposit on the surface during the reduction process. This leaching and redepositing processes continue and finally stop completely after 15 000 cycles of ADT confirming absence of any oxidizable Ge present at the surface (Figure 3d; Figure S15, Supporting Information). This signifies that surface reconstruction takes place during the HER conditions for first few cycles. This surface stabilizes the HER kinetics which is reflected in 500 h stability during CA. The dissoluted Ge in the electrolyte was verified by ICP-OES (Figure S7, Supporting Information) and Figure S16 and Note S1 (Supporting Information) explains that Ge dissolution exists as Ge cations in the electrolyte which when given fresh solution no more shows any Ge dissolution. Figure S17 (Supporting Information) shows the ECSA normalized LSV of 20% Pt/C and Pt_3Ge-(202) catalysts which shows Pt_3Ge-(202) is better with respect to specific activity. Post-HER X-ray photoelectron spectroscopy (XPS) provides a downshift of binding energy of Pt 4f edge (Figure 3e), which indicates that notable charge transfer from Ge to Pt takes place leading to Ge leaching and faster electron transfer from Pt to H^+. Further verification of conversion of dual-phase compound to single phase Pt_3Ge has been done by post-HER PXRD (Figure 3f). The control studies have been performed where the XRD data was taken on the catalyst which were exposed to a) oxidation potential for 100 activation CVs, b) 24 h CA without any activation (no oxidation potential experienced), and c) 500 h CA with activation. All these three conditions steer the catalyst to reach a similar fate, which is nothing but a total collapse of the "sacrificial phase" Pt_3Ge_2 and the operando generation of Pt_3Ge single phase with (202) exposed facet. Germanium leaching was observed even in the absence of oxidation potential which confirms the inherent charge transfer from Ge to Pt during HER.

3. Driving Forces for Accelerated HER

3.1. Charge Transfer

As mentioned in the previous section, the selective dissolution of Ge and accelerated hydrogen production have been originated through the charge transfer from Ge to Pt. To map it in depth, XAS (**Figure 4**a; Figures S18 and S19 and Table S5, Supporting Information) and XPS (Figure 4b; Figures S18, S20, and S21) techniques have been employed. Pt L_{III} edge X-ray absorption Near Edge spectra (XANES) represented in Figure 4a provide the oxidation state of Pt in the order of Pt foil < Pt_3Ge-(202) < Pt_3Ge-(110). This trend is well corroborated in the Pt 4f edge XPS spectra (Figure 4b). Pt_3Ge-(202)

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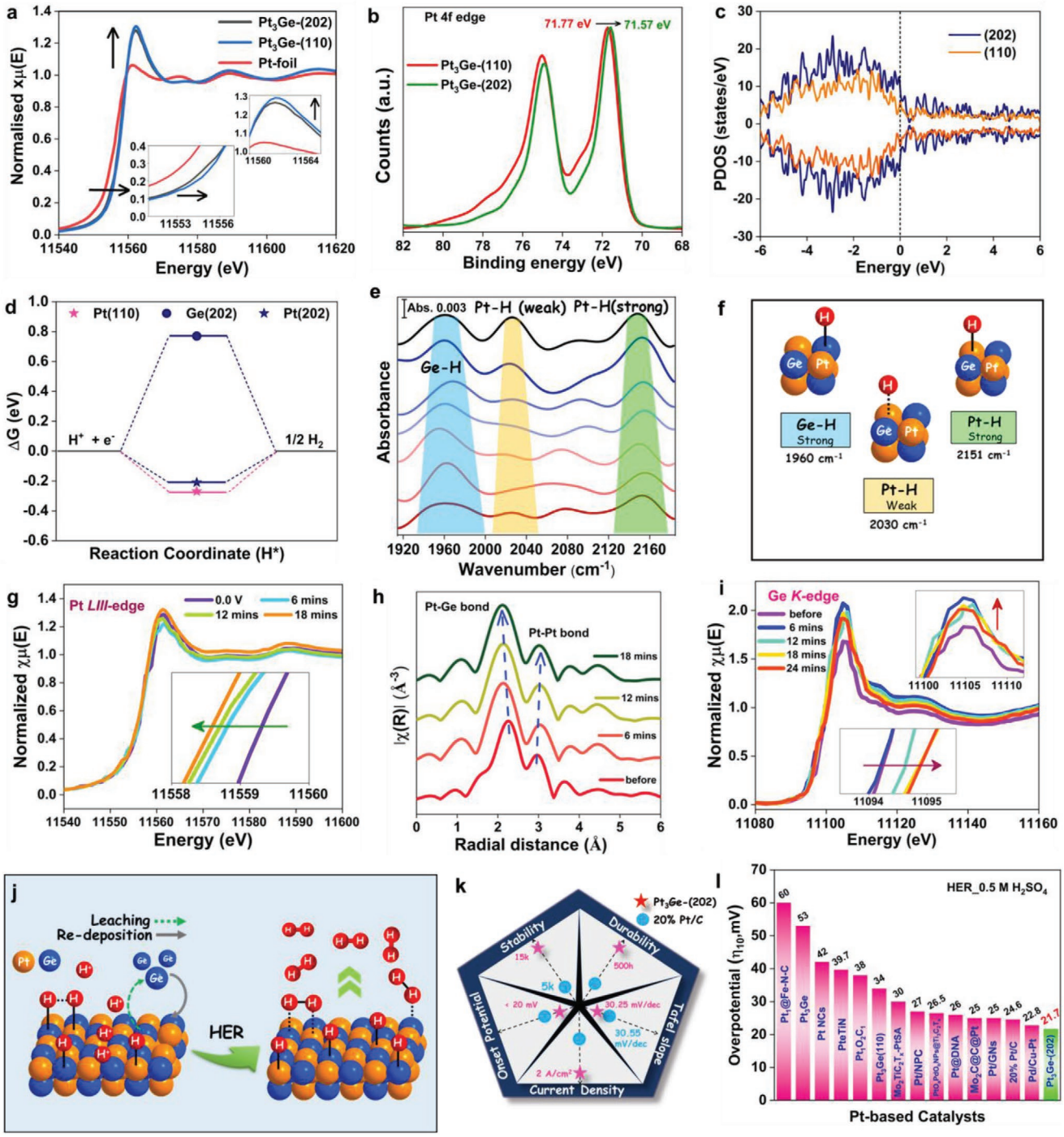

Figure 4. a) The normalized Pt L_{III} edge XANES spectra of Pt_3Ge-(110), Pt_3Ge-(202) and Pt foil. The insets show the zoomed-in images of the white line intensity and absorption edge. b) XPS of Pt 4f edge for Pt_3Ge-(202) and Pt_3Ge-(110). c) PDOS for Pt_3Ge-(202) and Pt_3Ge-(110). d) Energy profile diagram for H* adsorption on various sites of (110) and (202) facets of Pt_3Ge. e) In situ ATR FTIR for Pt_3Ge-(202) during CA for 30 min at –0.3 V vs RHE 0.5 M H_2SO_4. f) Different binding modes of hydrogen on Pt and Ge and their respective stretching frequencies. g) Pt *LIII*-edge XANES spectra during operando HER conditions. h) Fourier transform *R*-space data for Pt edge during the HER. i) Ge *K*-edge XANES spectra measured during the HER. j) Schematic for reaction mechanism derived from in situ IR. k) The comparison of Fab-five parameters of Pt_3Ge-(202) with commercial 20% Pt/C. l) Comparison (overpotential values at curent density of 10 mA cm^{-2}) of HER performance in acidic media of different Pt-based cataysts reported in the literature with that of Pt_3Ge-(202).[28,31,70–79]

XPS fitted spectra shows Pt has slightly high binding energy due to Pt–Ge charge separation and a slight amount of PtO present as surface oxide (Figure S18b, Supporting Information).[64] Lower oxidation state of Pt in Pt_3Ge-(202) depicts that the charge transfer (electron transfer)[67] from Ge to Pt is significant in the case of Pt_3Ge-(202) (Figure 4b). The electron transfer

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from Ge to Pt is playing a crucial role in enhancing the HER kinetics because the electron-donation to H^+ by Pt sites is significantly more than elemental Pt. The *R*-space data shows Pt–Pt and Pt–Ge bond lengths at relevant positions with no Pt-oxides present in the bulk (Figures S20 and S21, Supporting Information). The peak corresponding to Pt–Pt bond length is slightly shifted to a smaller distance than Pt foil indicating bond contraction, which is due to the observed higher oxidation state of Pt in Pt_3Ge than elemental Pt.[68] Pt–Pt and Pt–Ge bond distances in Pt_3Ge(202) are slightly larger than that in Pt_3Ge-(110) attributed to the less oxidation state of Pt in Pt_3Ge-(202) than Pt_3Ge-(110) (Table S5, Supporting Information). As obtained from XAFS, the coordination number of Ge around Pt is more in Pt_3Ge-(202) compared to Pt_3Ge-(110) which is justified from the fact that (202) facet contains both Pt and Ge atoms and facet (110) has only Pt atoms (Figure 1b; Table S5, Supporting Information). Figure S22 (Supporting Information) shows the Ge 3d XPS spectra before and after the HER, which shows that Ge sites are undergoing oxidation during the HER. This again confirms the charge transfer from Ge to Pt during the HER (Figure 3e shows about Pt 4f XPS).

3.2. Electronic Structure Tuning

In order to have a profound understanding of the electronic properties of the two facets, we have calculated their partial density of states (PDOS) along the projections on Pt and Ge sites. Structural models are shown in Figure S23 in the Supporting Information. In order to explore the thermodynamic stability of (202) facet with some Ge atoms removed, we have considered one of the most important cases of vacancy induced defect in the Pt_3Ge surface along (202) direction, while removing the terminating Ge atom from the surface along (202) direction.

After obtaining the minimum energy configuration with the lowest energy, we have determined the corresponding formation energy for the Ge vacancy induced defect in (202) facet of Pt_3Ge as 2.91 eV. Table S6 (Supporting Information) shows that the formation energy for (202) facet is negative value for the pristine phase where no Ge deficiency was created. This confirms that the Ge vacancy at the surface will lead to instability in the Pt_3Ge along its growth in (202) direction as compared to the pristine counterpart (without Ge deficiency) and therefore, we can theoretically support the fact that by increasing Ge content (Pt:Ge ration from 3:1 to 1:1) we get a stabilized (202) facet which is having a positive formation energy value when Ge vacancies are present in (202) facet. This confirms and establishes the formation mechanism of selective (202) facet formation during synthesis when more Ge were taken.[69] As we can observe from Figure S24 (Supporting Information), electronic property of both the facets is mainly governed by Pt. A careful observation (as in Figure 4c) reveals that near the Fermi region the states are higher in case of (202) plane compared to (110). This indicates the probability of having a greater number of electrons in the former case which showed better HER performance.

To conduct detailed analysis of facet-selective HER activity, free energy of H* adsorption on different sites of (110) and (202) facets were calculated (Figure 4d; Table S6, Supporting Information) illustrating H* adsorption on Pt of (110) and (202) are exothermic with DG_{H*} values of –0.27 and –0.21 eV, respectively. More optimum binding is expected on Pt atoms of (202) facet due to DG_{H*} close to 0 which leads to easy desorption of H_2 from the surface.[80] A DG_{H*} value of 0.77 eV for Ge site on (202) facet facilitates easy breaking of Ge–H bond leading to faster kinetics.

3.3. Mechanistic Pathway

The enhanced performance of Pt_3Ge-(202) compared to the commercial Pt/C can be explained if the mechanism of the HER can be mapped in operando condition. In situ attenuated total reflection-Fourier transform infrared spectroscopy (ATR-FTIR) measurements (Figure S25, Supporting Information) have been performed to probe Ge–H bonding[81] along with highly probable Pt–H bonding (Figure 4e; Figure S26, Supporting Information).[82,83] It is already reported in the literature that Pt–H[82,83] and Ge–H[49,81] bonds show a stretching frequency at 2030–2150 and 1960 cm^{-1}, respectively (Figure 4f). Pt_3Ge-(110) shows a prominent peak at 2030 cm^{-1} corresponding to Pt–H bond stretching vibration. Whereas, Pt_3Ge-(202) shows prominent peaks at 1960 and 2151 cm^{-1} corresponding to stretching vibrations of Ge–H and Pt–H bonds, respectively. Notably, peaks at 1960 cm^{-1} were totally absent in case of Pt_3Ge-(110) (Figure S26, Supporting Information) indicating that Ge atoms are not exposed in Pt_3Ge-(110), which is in perfect agreement with the crystal structure of Pt_3Ge (Figure 1b). This exposure of Ge in Pt_3Ge-(202) is the driving force for the accelerated hydrogen production as it contributes to H-adsorption along with Pt.

3.4. Probing Catalyst Structure during Reaction

Figure S27 (Supporting Information) shows the cell set-up for in situ XAS measurements. It is clearly observed from in-situ XANES spectra of Pt *LIII* edge that oxidation state of Pt is decreasing with increasing time during the HER (Figure 4g). *R*-space data for Pt *LIII* edge shows an interesting trend of increment of Pt–Pt bond distance and decrease of P–Ge bond distance (Figure 4h). Whereas the oxidation state for Ge edge is increasing with time during CA at a reducing potential (Figure 4i). We can infer the charge transfer from Ge to Pt from the XANES spectra. From close observation of the inset images of XANES spectra for both edges, it is found that the change in oxidation state is very prominent in first few minutes, and the change stopped after some time (after ≈12 min for Pt and 18 min for Ge), which indicates that the structural modifications occur at the first instant of the HER, then the structural reconstruction ceases and Ge leaching also stops after attaining a stable structure. *R*-space data for Pt *LIII* edge (Figure 4h) supports that Pt–Ge bond contraction occurs due to increase oxidation state of Ge and Ge leaching can also shorten the bond length. The operando XAS study finally confirms all other observations from electrochemical control experiments (Ge leaching, increasing activity after some time of CA),

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ICP-OES measurements showing no Pt leaching but only Ge leaching which stops after some time.[84] It is understood why the current density of our catalyst during a CA (both acidic and alkaline media) shows an enhancement of current density just after some time of CA. The schematic of the HER mechanism derived from the in situ ATR FTIR studies is shown in Figure 4j. Figure S28 (Supporting Information) shows how the catalyst retains its well dispersed spherical morphology, its size and also the exposed facet (202). This strongly suggests that the catalyst structure does not change after extensive HER and thus is highly stable. Figure S29 (Supporting Information) shows the wide-region HRTEM image indexing after the HER, showing that after the HER Pt_3Ge_2 phase completely destroyed and leaving behind only (202) facets of Pt_3Ge. Hydrogen quantitfication performed using the inverse burette method confirms the efficient production of hydrogen by Pt_3Ge-(202) (Figure S30, Supporting Information).

4. Conclusion

A new hydrogen production catalyst, Pt_3Ge-(202), has been discovered using a novel synthetic approach. To the best of our knowledge, this is the first ever material holding all Fabulous Five properties ((1) onset potential, (2) current density, (3) Tafel slope value, (4) stability in wide pH range, and (5) industrial-standard durability better than the state-of-the-art commercial Pt/C for the industrial-standard production of hydrogen (Figure 4k). The synthetically tuned Pt_3Ge-(202) surpasses the onset potential of Pt and excellent stability having much lower onset potential and faster kinetics compared to all other Pt-based catalysts reported in the literature (Figure 4l; Table S7, Supporting Information). Finally, the industrial production of hydrogen has been demonstrating by setting up a flow cell in alkaline media, which demonstrated that the catalyst Pt_3Ge-(202) shows long term durability for 75 h when done at 500 mA cm^{-2} current density. A comparison provided in Table S8 (Supporting Information) clearly confirm that the combination of all the Fab-Five parameters is indeed impossible in a single material and our catalyst stands out as the near-ideal material for industrial production of hydrogen. The mechanism and charge transfer kinetics of HER on Pt_3Ge-(202) is clearly manifested by in situ IR and XAS measurements and theoretical calculations.

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

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Conflict of Interest

The authors declare no conflict of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Keywords

electrochemistry, hydrogen production, intermetallics, water electrolysis

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